Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The sum of the rational terms in the expansion of $\left(\sqrt{2}+3^{1/5}\right)^{10}$ is

(A) 41

(B) 230

(C) 520

(D) none of these

Q.2

The greatest value of the term independent of x in the expansion of $\left(x\sin\alpha+x^{-1}\cos\alpha\right)^{10}$, $\alpha\in R$, is

(A) 2^5

(B) $\frac{10!}{(5!)^2}$

(C) $\frac{1}{2^5}\cdot\frac{10!}{(5!)^2}$

(D) none of these

Q.3

If the coefficients of rth and $(r+1)^{\text{th}}$ terms in the expansion of $(3+7x)^{29}$ are equal, then r equals

(A) 15

(B) 21

(C) 14

(D) none of these

Q.4

If n is an odd natural number, then $\sum_{r=0}^{n}\frac{(-1)^r}{{}^nC_r}$ equals

(A) 0

(B) $1/n$

(C) $n/2^n$

(D) none of these

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Q.5

If in the expansion of $(1+x)^m(1-x)^n$, the co-efficient of x and x^2 are 3 and -6 respectively, then m is

(A) 6

(B) 9

(C) 12

(D) 24

Q.6

In the expansion of the following expression $1+(1+x)+(1+x)^2+\ldots\ldots+(1+x)^n$ the coefficient of $x^k(0 \leq k \leq n)$ is

(A) ${}^{n+1}C_{k+1}$

(B) ${}^nC_k$

(C) ${}^nC_{n-k+1}$

(D) none of these

Q.7

If $(1+x)^n = C_0 + C_1x + C_2x^2 + \ldots\ldots + C_nx^n$, then $\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \ldots\ldots\ldots + \frac{nC_n}{C_{n-1}}$ is equal to

(A) $\frac{n(n-1)}{2}$

(B) $\frac{n(n+2)}{2}$

(C) $\frac{n(n+1)}{2}$

(D) $\frac{(n-1)(n-2)}{2}$

Q.8

$10^n + 3.4^{n+2} + 5, \forall n \in N$ is divisible by

(A) 7

(B) 4

(C) 9

(D) 15

Q.9

If the sum of the coefficients in the expansion of $(x+y)^n$ is 4096, then the greatest coefficient in the expansion is

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(A) 1594

(B) 792

(C) 924

(D) none of these

Q.10

If $\left(1-x+x^2\right)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_{2n} x^{2n}$, then $a_0 + a_2 + a_4 + \ldots\ldots + a_{2n}$ equals

(A) $\frac{3^n+1}{2}$

(B) $\frac{3^n-1}{2}$

(C) $\frac{1-3^n}{2}$

(D) $3^n + \frac{1}{2}$

Q.11

$x\left(x^{n-1} - na^{n-1}\right) + a^n(n-1)$ is divisible by $(x-a)^2$ for

(A) $n > 1$

(B) $n > 2$

(C) all $n \in N$

(D) none of these

Q.12

The larger of $99^{50} + 100^{50}$ and 101^{50} is

(A) $99^{50} + 100^{50}$

(B) both are equal

(C) 101^{50}

(D) none of these

Q.13

The coefficient of x^5 in the expansion of $\left(x^2 - x - 2\right)^5$ is

(A) -83

(B) -82

(C) -81

(D) 0

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Q.14

If $(1+x)^{15} = C_0 + C_1 x + C_2 x^2 + \ldots\ldots + C_{15} x^{15}$, then $C_2 + 2C_3 + 3C_4 + \ldots\ldots + 14C_{15}$ is equal to

(A) 14.2^{14}

(B) $13.2^{14} + 1$

(C) $13.2^{14} - 1$

(D) none of these

Q.15

Given that 4th term in the expansion of $\left(2 + \frac{3}{8}x\right)^{10}$ has the maximum numerical value, then the maximum possible range of value of x for which this will be true is given by

(A) $-\frac{64}{21} < x < -2$

(B) $-\frac{64}{21} < x < 2$

(C) $\frac{64}{21} < x < 4$

(D) none of these

Q.16

The sum $\sum_{i=0}^{m} \begin{pmatrix} 10 \\ i \end{pmatrix} \begin{pmatrix} 20 \\ m-i \end{pmatrix}$, (where $\begin{pmatrix} p \\ q \end{pmatrix} = 0$ if $p < q$) is maximum when m is

(A) 5

(B) 15

(C) 10

(D) 20

Q.17

If $(1+x)^n = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_n x^n$, then $\left(1 + \frac{a_1}{a_0}\right)\left(1 + \frac{a_2}{a_1}\right)\left(1 + \frac{a_3}{a_2}\right)\ldots\ldots\cdot\left(1 + \frac{a_n}{a_{n-1}}\right)$ is equal to

(A) $\frac{n^n}{n!}$

(B) $\frac{(n+1)^n}{n!}$

(C) $\frac{n^{n+1}}{(n+1)!}$

(D) none of these

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Q.18

If C_r stands for ${}^nC_r$, then $(C_0 + C_1) + (C_1 + C_2) + \ldots.. + (C_{n-1} + C_n)$ is equal to

(A) $2^n - 1$

(B) $2^{n+1} + 1$

(C) $2^{n+1} - 1$

(D) $2^{n+1} - 2$

Q.19

If $[x]$ denotes greatest integer $\leq x$ and $g = (3 + \sqrt{5})^n$, then $[g]$ must be

(A) $(3 + \sqrt{5})^n + (3 - \sqrt{5})^n + 1$

(B) $(3 + \sqrt{5})^n + (3 - \sqrt{5})^n - 1$

(C) $(3 + \sqrt{5})^n + (3 - \sqrt{5})^n$

(D) $(3 + \sqrt{5})^n - (3 - \sqrt{5})^n$

Q.20

$(n+2) \cdot {}^nC_0 \cdot 2^{n+1} - (n+1) \cdot {}^nC_1 \cdot 2^n + n \cdot {}^nC_2 \cdot 2^{n-1} - \ldots\ldots$ is equal to

(A) 4

(B) $4n$

(C) $4(n+1)$

(D) $2(n+2)$

Q.21

If $(1+x)^{10} = a_0 + a_1 x + a_2 x^2 + \ldots\ldots + a_{10} x^{10}$, then $(a_0 - a_2 + a_4 - a_6 + a_8 - a_{10})^2 + (a_1 - a_3 + a_5 - a_7 + a_9)^2$ is 2^k then $k =$

Q.22

The value of x in the expression $\left[x + x^{\log_{10} x}\right]^5$, if the third term in the expansion is $10,00,000$ is

Q.23

The greatest integer less than or equal to $(\sqrt{2} + 1)^6$ is

Q.24

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If in the expansion of $\left(x^3-\frac{1}{x^2}\right)^n, n \in N$, sum of the coefficient of x^5 and x^{10} is 0, then value of n is

Q.25

The coefficient of three successive terms in the expansion of $(1+x)^n$ are 165,330 and 462 respectively, then the value of n will be

Q.26

The number of terms in the expansion of $(2x+3y-4z)^n$ for $\mathrm{n}=10$ is

Q.27

The number of terms which are free from radical signs in the expansion of $\left(y^{1/5}+x^{1/10}\right)^{55}$ is

Q.28

In the polynomial $(x-1)(x-2)(x-3)\ldots\ldots(x-100)$, the coefficient of x^{99} is k then $|\mathrm{k}|=$

Q.29

Let $P(x)$ be a polynomial with integral coefficients. If there exist two integers a and b such that $P(a)-P(b)=1$, then $|\mathrm{a}-\mathrm{b}|=$

Q.30

If the last term in the binomial expansion of $\left(2^{1/3}-\frac{1}{\sqrt{2}}\right)^n$ is $\left(\frac{1}{3^{5/3}}\right)^{\log_3 8}$, then the 5th term from the beginning is

Answers & Solutions

Q.1 Answer:

41

Solution:

$T_{r+1}={}^{10}C_r(\sqrt{2})^{10-r}\left(3^{1/5}\right)^r$, **where r varies from 0 to 10 as there will be only 11 terms.**
In T_{r+1} the powers of 2 and $\frac{10-r}{2}$ and $\frac{r}{5}$ where $0 \leq r \leq 10$.
$\frac{r}{5}$ will be an integer for $r=0,5,10$ but $\frac{10-r}{2}$ will not be an integer for $r=5$.
Thus both powers are integers for $r=0$ and 10 .
Hence T_1 and T_{11} will have rational coefficients whose sum is
${}^{10}C_0(\sqrt{2})^{10} \cdot 1+{}^{10}C_{10}.1.3^2=32+9=41$

Q.2 Answer:

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$\frac{1}{2^5} \cdot \frac{10!}{(5!)^2}$

Solution:

$t_{r+1} = {}^{10}C_r (x \sin\alpha)^{10-r} \cdot \left(\frac{\cos\alpha}{x}\right)^r$

It is independent of x if $r = 5$.

$\therefore$ the term independent of x $= {}^{10}C_5 \cdot \sin^5\alpha \cdot \cos^5\alpha = {}^{10}C_5 \cdot \frac{1}{2^5}(\sin 2\alpha)^5 \leq {}^{10}C_5 \cdot \frac{1}{2^5}$

Q.3 Answer:

21

Solution:

We have $T_{r+1} = {}^{29}C_r 3^{29-r}(7x)^r = \left({}^{29}C_r \times 3^{29-r} \times 7^r\right) x^r$

$\therefore$ $a_r =$ **coefficient of** $(r+1)^{\text{th}}$ **term** $= {}^{29}\text{C}_\text{r} \times 3^{29-r} \times 7^r$

and $a_{r-1} =$ **coefficient of** r^{th} **term** $= {}^{29}C_{r-1} \times 3^{30-r} \times 7^{r-1}$

Now, $a_r = a_{r-1}$

$\Rightarrow$ $\quad {}^{29}C_r \times 3^{29-r} \times 7^r = {}^{29}C_{r-1} \times 3^{30-r} \times 7^{r-1}$

$\Rightarrow \frac{{}^{29}C_r}{{}^{29}C_{r-1}} = \frac{3}{7} \quad \Rightarrow \quad \frac{30-r}{r} = \frac{3}{7} \Rightarrow r = 21.$

Q.4 Answer:

0

Solution:

$\sum_{r=0}^{n} \frac{(-1)^r}{{}^nC_r} = \sum_{r=0}^{\frac{n+1}{2}} \left[\frac{(-1)^r}{{}^nC_r} + \frac{(-1)^{n-r}}{{}^nC_{n-r}}\right] = \sum_{r=0}^{\frac{n+1}{2}} (-1)^r \left[\frac{1}{{}^nC_r} + \frac{(-1)^n}{{}^nC_{n-r}}\right]$

$= \sum_{r=0}^{\frac{n+1}{2}} \left[\frac{1}{{}^nC_r} - \frac{1}{{}^nC_r}\right]$

$[Qn$ **is odd and** ${}^nC_r = {}^nC_{n-r}]$

Q.5 Answer:

12

Solution:

$(1+x)^m(1-x)^n = \left(1 + mx + \frac{m(m-1)}{2}x^2 + \ldots..\right)\left(1 - nx + \frac{n(n-1)}{2}x^2 + \ldots...\right)$

Co-efficient of $x = (m-n) = 3$ **(given)**

Co-efficient of $x^2 = \frac{m(m-1)}{2} - mn + \frac{n(n-1)}{2} = -6$ **(given)**

$\Rightarrow m^2 - m - 2mn + n^2 - n = -12$

i.e. $m^2 + n^2 - 2mn - (m+n) = -12$

i.e. $(m-n)^2 - (m+n) = -12$**...(i)**

Also $m - n = 3$ **...(ii)**

By (i) and (ii)

$\Rightarrow 9 - (m+n) = -12 \quad \therefore \quad m+n = 21 \quad \ldots(iii)$

(ii) + (iii) gives $2m = 24 \quad \therefore m = 12$

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Q.6 Answer:

${}^{n+1}C_{k+1}$

Solution:

The expression being in G.P.

$E = 1 + (1+x) + (1+x)^2 + \ldots\ldots + (1+x)^n$

$\frac{(1+x)^{n+1}-1}{(1+x)-1} = x^{-1}\left\{(1+x)^{n+1} - 1\right\}$

$\therefore$ **The coefficient of** x^k **in** E **The coefficient of** x^{k+1} **in** $\left\{(1+x)^{n+1} - 1\right\} = {}^{n+1}C_{k+1}$

Q.7 Answer:

$\frac{n(n+1)}{2}$

Solution:

$\frac{C_1}{C_0} + \frac{2C_2}{C_1} + \frac{3C_3}{C_2} + \ldots\ldots + \frac{nC_n}{C_{n-1}}$

$= \frac{n}{1} + 2\frac{\frac{n(n-1)}{1.2}}{n} + 3\frac{\frac{n(n-1)(n-2)}{3.2.1}}{\frac{n(n-1)}{1.2}} + \ldots\ldots + n \cdot \frac{1}{n}$

$= n + (n-1) + (n-2)\ldots.. + 1 = \sum n = \frac{n(n+1)}{2}$

Q.8 Answer:

9

Solution:

$P(n) = 10^n + 3.4^{n+2} + 5$ **is divisible by** 9.

For $n = 1$,

$P(1) = 10^1 + 3 \cdot 4^{1+2} + 5$

$= 10 + 3 \times 4^3 + 5$

$= 10 + 192 + 5$

$= 207$ **is divisible by** 9.

Hence, the given statement is true for $n = 1$,

i.e, $\mathrm{P}(1)$ **is true**

Let the statement $\mathrm{P}(\mathrm{m})$ **is true i.e,**

$P(m) = 10^m + 3 \cdot 4^{m+2} + 5$ **is divisible bo** 9. $10^m + 3 \cdot 4^{m+2} + 5 = 9m($ **Let** $)$

Now, we have to prove that the given statement is true for $n = m+1$**, i.e.,** $P(m+1)$ **is true.**

$$\text{Now } 10^{(k+t)} + 3 \cdot 4(k+1) + 2 + 5$$

$$P(m+1) = 10^{m+1} + 3 \cdot 4^{(m-1)+2+5}$$

$$= 10^{m \cdot} 10 + 3 \cdot 4^{m+2} \cdot 4 + 5$$

$= \left(9m - 3 \cdot 4^{m+2} - 5\right) \cdot 10 + 3 \cdot 4^{m+2 \cdot 4} + 5$

$= 9.10m - 30 \cdot 4^{m+2} - 50 + 12 \cdot 4^{m+2} + 5$

$$= 9.10m - 18 \cdot 4^{m+2} - 45$$

$= 9\left[10m - 2 \cdot 4^{m+2} - 5\right]$ is divisible by 9.

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Q.9 Answer:

924

Solution:

Putting $x = y = 1$, the sum of the coefficients is $2^n = 4096 = 16^3 = \left(2^4\right)^3$
$\therefore \quad n = 12$
If ${}^nC_r = {}^{12}C_r$ will be greatest when $r = \frac{n}{2} = 6$, as n is even.
Hence ${}^{12}C_6 = 924$

Q.10 Answer:

$\frac{3^n+1}{2}$

Solution:

Put $x = 1, -1$ successively in the given equation and add.

Q.11 Answer:

all $n \in N$

Solution:

For $n = 1$, the given expression is zero and hence the statement is true for $n = 1$
Suppose the statement is true for some m
$\therefore x\left(x^{m-1} - ma^{m-1}\right) + (m-1)a^m = (x-a)^2 Q$
$\therefore$ For $n = m+1$, $\left(x^m - (m+1)a^m\right) + ma^{m+1}$
$= (x-a)^2 xQ + ma^{m-1}\left(x^2 - 2ax + a^2\right) = (x-a)^2\left[xQ + ma^{m-1}\right]$
$\therefore$ By induction, the statement is true for each n.

Q.12 Answer:

101^{50}

Solution:

We have. $101^{50} = (100+1)^{50} = 100^{50} + 50.100^{49} + \frac{50.49}{2.1}100^{48} + \ldots\ldots$ **...(i)**
and $99^{50} = (100-1)^{50} = 100^{50} - 50.100^{49} + \frac{50.49}{2.1}100^{48}$ **...(ii)**
Subtracting (ii) from (i), we get
$101^{50} - 99^{50} = 2 \times 100^{50} + 2\frac{50.49.48}{1.2.3}100^{47} > 100^{50}$
Hence $101^{50} > 100^{50} + 99^{50}$

Q.13 Answer:

-81

Solution:

$\left(x^2 - x - 2\right)^5 = (x-2)^5(1+x)^5$
$= \left[{}^5C_0 x^5 - {}^5C_1 x^4 \times 2 + \ldots.\right]\left[{}^5C_0 + {}^5C_1 x + \ldots\ldots\right]$

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Coefficient of x^5 in expansion

$= \left({}^5C_0\right)^2 - 2\left({}^5C_1\right)^2 + 2^2\left({}^5C_2\right)^2 - 2^3\left({}^5C_3\right)^3 + 2^4\left({}^5C_4\right)^2 - 2^5\left({}^5C_5\right)^2 = -81$

Q.14 Answer:

$13.2^{14} + 1$

Solution:

We have $(1+x)^{15} = C_0 + C_1x + C_2x^2 + \ldots\ldots + C_{15}x^{15}$

$\Rightarrow \quad \frac{(1+x)^{15}-1}{x} = C_1 + C_2x + \ldots\ldots + C_{15}x^{14}$

Differentiating both sides with respect to x**, we get** $= \frac{x.15(1+x)^{14}-(1+x)^{15}+1}{x^2} = C_2 + 2C_3x + \ldots\ldots + {}^{14}C_{15}x^{13}$

Putting $x = 1$**, we get** $C_2 + 2C_3 + \ldots.. + 14C_{15} = 15.2^{14} - 2^{15} + 1 = 13.2^{14} + 1$

Q.15 Answer:

$-\frac{64}{21} < x < -2$

Solution:

T_3, T_4, T_5 **in the given expansion are respectively**

${}^{10}C_2 2^8\left(\frac{3x}{8}\right)^2, {}^{10}C_3 2^7\left(\frac{3x}{8}\right)^3, {}^{10}C_4 2^6\left(\frac{3x}{8}\right)^4$

Or $1620x^2, 810x^3, \frac{8505}{32}x^4$

Given that $|T_4| > |T_3|$ **and** $|T_4|_> |T_5|$

$\therefore \quad |x| > 2 \text{ and } \frac{64}{21} > |x|$

$2 < |x| < \frac{64}{21}$ **...(i)**

The above inequality (i) is equivalent to two inequalities

$2 < x < \frac{64}{21} \text{ and } -\frac{64}{21} < x < -2$

Q.16 Answer:

15

Solution:

$(1+x)^{10} = {}^{10}C_0 + {}^{10}C_1x + \ldots\ldots + {}^{10}C_{10}x^{10}$

$(1+x)^{20} = {}^{20}C_0 + {}^{20}C_1x + \ldots\ldots {}^{20}C_{m-i}x^{m-i} + \ldots.{}^{20}C_{20}x^{20}$

$\sum {}^{10}C_i {}^{20}C_{m-i} =$ **coefficient** x^m **in** $(1+x)^{30}$

$= {}^{30}C_m$**, which is maximum if** $m = 15$

Q.17 Answer:

$\frac{(n+1)^n}{n!}$

Solution:

Clearly $a_r = {}^nC_r \quad \Rightarrow \quad \frac{a_r}{a_{r-1}} = \frac{{}^nC_r}{{}^nC_{r-1}} = \frac{(n-r+1)}{r} \quad \Rightarrow \quad 1 + \frac{a_r}{a_{r-1}} = \frac{n+1}{r}$

$\Rightarrow {}^n_{r=1}\left(1 + \frac{a_r}{a_{r-1}}\right) = \prod_{r=1}^n \frac{(n+1)}{r} = \frac{(n+1)^n}{n!}$

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Q.18 Answer:

$2^{n+1} - 2$

Solution:

$$(C_0 + C_1) + (C_1 + C_2) + \ldots + (C_{n-1} + C_n)$$
$$= \binom{n+1}{1}C_1 + \binom{n+1}{2}C_2 + \ldots + \binom{n+1}{n}C_n$$
$$= {}^{n+1}C_0 + {}^{n+1}C_1 + {}^{n+1}C_2 + \ldots + {}^{n+1}C_n + {}^{n+1}C_{n+1} - 2$$
$$= 2^{n+1} - 2$$

Q.19 Answer:

$(3+\sqrt{5})^n + (3-\sqrt{5})^n - 1$

Solution:

$g = (3+\sqrt{5})^n = [g] + f$

Consider another fraction $f' = (3-\sqrt{5})^n$

$\Rightarrow \quad [g] + f + f' = (3+\sqrt{5})^n(3-\sqrt{5})^n =$ **integer**

and $0 < f < 1$ **and** $0 < f' < 1 \Rightarrow f + f' = 1$

So $[g] = (3+\sqrt{5})^n + (3-\sqrt{5})^n - 1$

Q.20 Answer:

$4(n+1)$

Solution:

$$t_{r+1} = (-1)^r \cdot (n-r+2)^n C_r \cdot 2^{n-r+1}$$
$$= (n+2)\cdot 2^{n+1}\cdot(-1)^r \cdot {}^nC_r \cdot \left(\tfrac{1}{2}\right)^r - 2^{n+1}\cdot(-1)^r \cdot r \cdot {}^nC_r \cdot \left(\tfrac{1}{2}\right)^r$$
$$= (n+2)\cdot 2^{n+1}\cdot(-1)^r \cdot {}^nC_r \cdot \left(\tfrac{1}{2}\right)^r - 2^n \cdot n \cdot (-1)^r \cdot {}^{n-1}C_{r-1} \cdot \left(\tfrac{1}{2}\right)^{r-1}$$
$$\therefore \quad \textbf{sum} = (n+2)\cdot 2^{n+1}\cdot\left\{{}^nC_0 - {}^nC_1\cdot\tfrac{1}{2} + {}^nC_2\cdot\left(\tfrac{1}{2}\right)^2 - \ldots..\right\} - n\cdot$$
$$2^n\left\{-{}^{n-1}C_0 + {}^{n-1}C_1\cdot\tfrac{1}{2} - {}^{n-1}C_2\cdot\left(\tfrac{1}{2}\right)^2 + \ldots\right\}$$
$$= (n+2)\cdot 2^{n+1}\cdot\left(1-\tfrac{1}{2}\right)^n + n\cdot 2^n\cdot\left(1-\tfrac{1}{2}\right)^{n-1} = 2(n+2) + 2n = 4n+4.$$

Q.21 Answer:

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Solution:

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Here $a_i = {}^{10}C_i$

ie, $(a_0 - a_2 + a_4 - a_6 + a_8 - a_{10})^2 +$
$(a_1 - a_3 + a_5 - a_7 + a_9)^2$
$= ({}^{10}C_0 - {}^{10}C_2 + {}^{10}C_4 - {}^{10}C_6 + {}^{10}C_8 - {}^{10}C_{10})^2 +$
$({}^{10}C_1 - {}^{10}C_3 + {}^{10}C_5 - {}^{10}C_7 + {}^{10}C_9)^2$
$\Rightarrow (({}^{10}C_0 - {}^{10}C_{10}) + ({}^{10}C_8 - {}^{10}C_2) + ({}^{10}C_4 - {}^{10}C_6))^2 +$
$\left((-1)(-2)^{\frac{10}{2}}\right)^2 = 2^{10}$

Since $\sum_{i=0}^{[n/2]n} C_{2i} = \begin{cases} 0, & \text{If } \frac{n+2}{4} \in \text{Integers} \\ (-1)^{[(n+2)/4)} & [l/2] \end{cases}$ and

$\sum_{i=0}^{[n/2]n} C_{2i+1} = \begin{cases} 0, & \text{If } \frac{n}{4} \in \text{Integers} \\ (-1)^{[n/4]} 2^{[n/2]} \end{cases}$

Option A is correct

Q.22 Answer:

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Solution:

$T_3 = {}^5C_2 \cdot x^2 \left(x^{\log_{10} x}\right)^3 = 10^6$
Put ${}^5C_2 = 10 \quad [\text{Q} \log_{10} 10 = 1]$
If $x = 10$, then $10^3 \cdot 10^2 = 10^5$ is satisfied $x = 10$

Q.23 Answer:

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Solution:

Let $(\sqrt{2}+1)^6 = I + F$, where I is an integer and $0 \le F < 1$
Let $f = (\sqrt{2}-1)^6$
$\Rightarrow \quad \sqrt{2} - 1 = \frac{1}{\sqrt{2}+1} \Rightarrow 0 < \sqrt{2} - 1 < 1 \Rightarrow 0 < f < 1$
Also, $I + F + f = (\sqrt{2}+1)^6 + (\sqrt{2}-1)^6$
$= 2\left[{}^6C_0 \cdot 2^3 + {}^6C_2 \cdot 2^2 + {}^6C_4 \cdot 2 + {}^6C_6\right] = 2(8 + 60 + 30 + 1) = 198$
Hence $F + f = 198 - I$ is an integer.
But $0 < F + f < 2$
Therefore, $F + f = 1$ and thus $I = 197$

Q.24 Answer:

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Solution:

$\left(x^3 - \frac{1}{x^2}\right)^n$
$(r+1)$ th term in the expansion of (x) is
${}^nC_r \left(x^3\right)^{n-r} \left(-\frac{1}{x^2}\right)^r$

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$= {}^nC_r(-1)^r x^{3n-5r}$

For coefficient of x^5, we set

$3n-5r=5 \Rightarrow r=\frac{3n-5}{5}=p$ **(say) and for coefficient of** x^{10}**, we set**

$3n-5r=10 \Rightarrow r=\frac{3n-10}{5}=q$ **(say)**

Note that $p-q=1$**. We are given** ${}^nC_p(-1)^p+{}^nC_q(-1)^q=0$

$\Rightarrow \quad {}^nC_p(-1)^p+{}^nC_{p-1}(-1)^{p-1}=0 \Rightarrow \quad {}^nC_p={}^nC_{p-1} \Rightarrow n-p=p-1$

$\Rightarrow \quad n=2p-1 \Rightarrow p=\frac{n+1}{2}$

Thus, $\frac{n+1}{2}=\frac{3n-5}{5} \Rightarrow 5n+5=6n-10 \quad \Rightarrow 15=n$

Q.25 Answer:

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Solution:

Let the coefficient of three consecutive term i.e., $(r+1)th, (r+2)th, (r+3)th$ **in expansion of** $(1+x)^n$ **are 165,330 and 462 respectively, then coefficient of** $(r+1)th$ **term** $= {}^nC_r=165$

Coefficient of $(r+2)th$ **term** $= {}^nC_{r+1}=330$ **and Coefficient of** $(r+3)th$ **term** $= {}^nC_{r+2}=462$

$\frac{{}^nC_{r+1}}{{}^nC_r}=\frac{n-r}{r+1}=2$ **or** $n-r=2(r+1)$ **or** $r=\frac{1}{3}(n-2)$

and $\frac{{}^nC_{r+2}}{{}^nC_{r+1}}=\frac{n-r-1}{r+2}=\frac{231}{165}=\frac{7}{5} \Rightarrow r=\frac{1}{12}(5n-19)$

$\Rightarrow \quad \frac{1}{3}(n-2)=\frac{1}{12}(5n-19) \Rightarrow n=11$

Q.26 Answer:

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Solution:

We have, $(2x+3y-4z)^n=\{2+(3-4)\}^n$

$={}^nC_0(2x)^n(3y-4z)^0+{}^nC_1(2x)^{n-1}(3y-4z)^1+{}^nC_2(2x)^{n-2}(3y-4z)^2+$
$\ldots {}^nC_{n-1}(2x)^1(3y-4z)^{n-1}+{}^nC_n(3y-4z)^n$

Clearly, the first term in the above expansion gives one term, second term gives two terms, third term gives three terms and so on.

So, Total number of term $=1+2+3+\ldots+n+(n+1)=\frac{(n+1)(n+2)}{2}$

Q.27 Answer:

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Solution:

The general term in the expansion of $\left(y^{1/5}+x^{1/10}\right)^{55}$ **is**

$T_{r+1}={}^{55}C_r\left(y^{1/5}\right)^{55-r}\left(x^{1/10}\right)^r={}^{55}C_r y^{11-r/5}x^{r/10}$

Clearly, T_{r+1} **will be independent of radical signs if** $r/5$ **and** $r/10$ **are integers, where** $0 \le r \le 55$**.**

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$\therefore \quad r = 0, 10, 20, 30, 40, 50$

Hence, there are 6 terms in the expansion of the given expression which are independent of radicals.

Q.28 Answer:

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Solution:

$(x-1)(x-2)(x-3)\ldots..(x-100)$

Number of terms $= 100;$

$\therefore$ **Coefficient of** $x^{99}(x-1)(x-2)(x-3)\ldots\ldots(x-100)$

$= (-1-2-3-\ldots..-100) = -(1+2+\ldots..+100)$

$= -\frac{100\times101}{2} = -5050$

Q.29 Answer:

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Solution:

Let $P(x) = a_0x^n + a_1x^{n-1} + \ldots\ldots + a_{n-1}x + a_n$

where $a_0, a_1, \ldots\ldots, a_n$ **are integers.**

Now, $1 = P(a) - P(b) = a_0\left(a^n - b^n\right) + a_1\left(a^{n-1} - b^{n-1}\right) + a_2\left(a^{n-2} - b^{n-2}\right) +$

$\ldots.. + a_{n-1}(a-b)$

$= (a-b)m$

where $m = a_0\left(a^{n-1} + a^{n-2}b + \ldots\ldots + ab^{n-2} + b^{n-1}\right) +$

$a_1\left(a^{n-2} + a^{n-3}b + \ldots\ldots + ab^{n-3} + b^{n-2}\right)$

$+ \ldots\ldots + a_{n-2}(a+b) + a_{n-1}$

is an integer.

As product of two integers is 1, either $a-b=1$ **and** $m=1$ **or** $a-b=-1$ **and** $m=-1$

$\therefore a$ **and** b **must be two consecutive integers i.e.** $|a-b| = 1$

Q.30 Answer:

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Solution:

Last term of $\left(2^{1/3} - \frac{1}{\sqrt{2}}\right)^n$ **is** $t_{n+1} = {}^nC_n\left(2^{1/3}\right)^{n-n}\left(-\frac{1}{\sqrt{2}}\right)^n$

$= {}^nC_n(-1)^n\frac{1}{2^{n/2}} = \frac{(-1)^n}{2^{n/2}}$

Also, $\left(\frac{1}{3^{5/3}}\right)^{\log_3 8} = \frac{1}{\left(3^{5/3}\right)^{3\log_3 2}} = 3^{-(5/3)\log_3 2^3} = 2^{-5}$

Thus, $\frac{(-1)^n}{2^{n/2}} = 2^{-5} \Rightarrow \frac{(-1)^n}{2^{n/2}} = \frac{(-1)^{10}}{2^5} \Rightarrow \frac{n}{2} = 5 \quad \Rightarrow n = 10$

Now, $t_5 = t_{4+1} = {}^{10}C_4\left(2^{1/3}\right)^{10-4}\left(-\frac{1}{\sqrt{2}}\right)^4$

$= \frac{10!}{4!6!}\left(2^{1/3}\right)^6(-1)^4\left(2^{-1/2}\right)^4 = 210\left(2^2\right)(1)\left(2^{-2}\right) = 210$

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